EC5.407 Wireless Communications — Assignment 2

Detection and Diversity in AWGN and Fading Channels

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1 Simulation set-up (common to all problems)

1.1 Model and conventions

All three problems use the conventions of the "Getting Started" section of the assignment. The constellation has unit average energy, so the SNR is set only by the noise variance:

$$E_s = \mathbb{E}|x|^2 = \frac{1}{M}\sum_{i=1}^{M}|u_i|^2 = 1, \qquad \mathbb{E}|w|^2 = \frac{1}{\mathrm{SNR}}, \qquad \mathrm{SNR} = \frac{E_s}{\mathbb{E}|w|^2}, \qquad \mathrm{SNR} = 10^{\mathrm{SNR_{dB}}/10}.$$

- **Constellations:** BPSK $\{+1, -1\}$ and 4-QAM $\{\pm A \pm jA\}$ with $A = 1/\sqrt{2}$. Both have unit energy, so they are compared at equal symbol energy.
- **Noise:** $w = \frac{1}{\sqrt{2\mathrm{SNR}}}(w_I + jw_Q) \sim \mathcal{CN}(0, 1/\mathrm{SNR})$, with $w_I, w_Q \sim \mathcal{N}(0, 1)$ independent.
- **Rayleigh fading:** $h = \frac{1}{\sqrt{2}}(h_I + jh_Q) \sim \mathcal{CN}(0, 1)$, a new value for every symbol (for every Alamouti block in Problem 3), known at the receiver.
- **Monte Carlo size:** $N = 10^6$ symbols per SNR point, for every curve and every L.
- **Seed:** set once at the start of every script with `np.random.default_rng(2026)`.
- **Vectorised:** all N symbols of an SNR point are processed at once. The only loops run over SNR points, schemes and L.

1.2 Code structure

Every block of the link in Fig. 1 of the assignment is written once, in `common/`, and every problem imports it. The code is Python 3.12 with NumPy, SciPy (only `erfc`) and Matplotlib. No library function for modulation, detection or error-rate calculation is used.

File	What it does
`constellation.py`	BPSK and 4-QAM, normalised to $E_s = 1$.
`channel.py`	Complex Gaussian samples; AWGN, Rayleigh, $1 \times L$ SIMO and 2×1 MISO channels.
`detector.py`	One ML (minimum-distance) detector for any constellation, $\hat{\imath} = \arg\min_m \lvert r - g\,u_m\rvert^2$, computed for all N samples at once as an $N \times M$ distance array.
`combining.py`	Receive combiners (none, MRC, SC) that turn the received samples into one scalar decision statistic.
`simulation.py`	Algorithm 1: draw indices, send through the channel, combine, detect, and count errors by comparing indices, $n_{\mathrm{err}} = \sum_n \mathbf{1}\{\hat{\imath}_n \neq i_n\}$.
`theory.py`	$Q(a) = \frac{1}{2}\,\mathrm{erfc}(a/\sqrt{2})$ as its own function, the formulas of the assignment table, and the exact 4-QAM SER of every scheme (Sec. 1.4).
`plotting.py` `analysis.py`	Figures (only points with $n_{\mathrm{err}} \geq 10$), CSV tables, SNR gaps, slopes and z-scores.
`problem1/ to problem3/`	One script per task. Each writes its figures, CSV table and a `summary.txt` with every number quoted below to `results/q1` – `results/q3`.
`overall_check.py`	Compares every plotted point of all problems with its exact theory (Sec. 1.5).

One detector for every receiver. Every receiver in this assignment reduces what it receives to an equivalent scalar channel

$$r = g\,x + n, \qquad n \sim \mathcal{CN}(0, 1/\mathrm{SNR}), \qquad g \text{ known at the receiver.} \tag{1}$$

The ML detector of Problem 1 is then applied without change; only the gain g differs: $g = 1$ (AWGN), $g = h$ (fading), $g = \|\mathbf{h}\|$ (MRC and beamforming), $g = h_{\ell^\star}$ (SC) and $g = \|\mathbf{h}\|/\sqrt{2}$ (Alamouti). The MRC and Alamouti outputs are divided by $\|\mathbf{h}\|$ so that the noise variance is $1/\mathrm{SNR}$ in every case. Without this division the gains would be $\|\mathbf{h}\|^2$ and $\|\mathbf{h}\|^2/\sqrt{2}$, with noise variance $\|\mathbf{h}\|^2/\mathrm{SNR}$. The SNR and the decisions would be the same (Sec. 3.1).

Reproducing. Each script is run as a module from the code root, for example `python3 -m problem2.receive_diversity`; the README lists all commands. Each script takes under a minute and reproduces its figures and numbers exactly.

1.3 How accurate is a simulated point?

A simulated point is $\hat{P}_\mathrm{e} = n_\mathrm{err}/N$. Each symbol is an independent Bernoulli(P_e) trial, so $\mathrm{std}(\hat{P}_\mathrm{e}) = \sqrt{P_\mathrm{e}(1-P_\mathrm{e})/N}$, and the relative accuracy is about $1/\sqrt{n_\mathrm{err}}$:

errors observed, n_err	10	100	1000	10 000
relative accuracy	±32%	±10%	±3%	±1%

A difference between simulation and theory is measured in units of this standard deviation,

$$z = \frac{\hat{P}_\mathrm{e} - P_\mathrm{e}}{\sqrt{P_\mathrm{e}(1-P_\mathrm{e})/N}}. \tag{2}$$

If the simulation is correct, z behaves like a standard normal variable. Most points then have $|z| < 2$, and the worst of a few hundred points is close to 3.

1.4 Exact 4-QAM theory used for checking

The assignment table gives an exact formula for BPSK in fading but only an approximation for 4-QAM, and no formula for Problems 2 and 3. To check every curve against an exact value, I compute the exact 4-QAM SER of each scheme in three steps.

1. **Every scheme is an AWGN channel with a random gain.** By (1), the SNR after detection is $\gamma\,\mathrm{SNR}$ with $\gamma = |g|^2$ (table below).
2. **Error probability for a fixed gain.** 4-QAM is two independent BPSK decisions (I and Q), each with half the energy: $P_\mathrm{e}(\gamma) = 2Q\big(\sqrt{\gamma\mathrm{SNR}}\big) - Q^2\big(\sqrt{\gamma\mathrm{SNR}}\big)$.
3. **Average over the gain.** Craig's forms of the Q-function,

$$Q(x) = \frac{1}{\pi}\int_0^{\pi/2} e^{-x^2/(2\sin^2\theta)}\,d\theta, \qquad Q^2(x) = \frac{1}{\pi}\int_0^{\pi/4} e^{-x^2/(2\sin^2\theta)}\,d\theta,$$

turn the average into one that only needs the moment-generating function $M_\gamma(s) = \mathbb{E}[e^{-s\gamma}]$:

$$P_\mathrm{e} = \frac{2}{\pi}\int_0^{\pi/2} M_\gamma\Big(\frac{\mathrm{SNR}}{2\sin^2\theta}\Big)d\theta - \frac{1}{\pi}\int_0^{\pi/4} M_\gamma\Big(\frac{\mathrm{SNR}}{2\sin^2\theta}\Big)d\theta. \tag{3}$$

Scheme	Gain $\gamma = \lvert g\rvert^2$	Distribution	$M_\gamma(s)$
Single antenna	$\lvert h\rvert^2$	Exp(1)	$1/(1+s)$
MRC (L), beamforming ($L=2$)	$\lVert\mathbf{h}\rVert^2$	Gamma$(L,1)$	$(1+s)^{-L}$
SC (L)	$\max_\ell \lvert h_\ell\rvert^2$	same as $\sum_{k=1}^{L} E_k/k$, $E_k \sim$ Exp(1)	$\prod_{k=1}^{L} k/(k+s)$
Alamouti	$\lVert\mathbf{h}\rVert^2/2$	$\frac{1}{2}$ Gamma$(2,1)$	$(1+s/2)^{-2}$

The integrals in (3) are evaluated with 256-point Gauss–Legendre quadrature (`ser_qam4_fading_mgf` in `theory.py`). I checked them in two ways:

- For one antenna, the result equals the closed form derived the same way, $P_e = (1-\mu) - \frac{1}{4}\left[1 - \frac{4}{\pi}\,\mu \arctan(1/\mu)\right]$ with $\mu = \sqrt{(\mathrm{SNR}/2)/(1+\mathrm{SNR}/2)}$.
- For MRC and SC with $L = 2, 4$, the result agrees with direct numerical integration over the density of γ to machine precision.

1.5 Overall check of all simulations

The script `overall_check.py` compares every plotted point ($n_{\mathrm{err}} \geq 10$) of Problems 1–3 with its exact value p, using the z-score (2), written with error counts as

$$z = \frac{n_{\mathrm{err}} - Np}{\sqrt{Np(1-p)}}.$$

If the simulation is unbiased, z behaves like a standard normal variable.

Three curves use exactly the same random numbers: MRC and SC with $L = 1$ (Problem 2), and the single-antenna curve of Problem 3. Both scripts start from seed 2026 and draw the symbols, channels and noise in the same order. Their error counts are identical, so they are counted *once*. This leaves 160 independent points (Table 1).

Table 1: Agreement between every simulated curve and its exact theory (`results/overall_check.txt`). "Max $|z|$" is the largest deviation over the plotted points of the curve. * This row also stands for SC, $L = 1$ and the single-antenna curve of Problem 3 (same random numbers, identical error counts).

Problem	Curve	Points	Max $\lvert z\rvert$	Problem	Curve	Points	Max $\lvert z\rvert$
1	BPSK, AWGN	10	2.59	2	MRC, $L = 4$	8	2.01
1	4-QAM, AWGN	13	1.95	2	SC, $L = 2$	13	1.64
1	BPSK, fading	21	3.02	2	SC, $L = 4$	9	1.12
1	4-QAM, fading	21	1.78	3	Beamforming	13	2.06
2	MRC, $L = 1$ *	13	1.85	3	Alamouti	13	2.55
2	MRC, $L = 2$	13	1.03	3	MRC, $L = 2$	13	1.85

Table 2 compares the statistics of all 160 z-values with those of a standard normal variable.

Table 2: Statistics of z over all 160 independent plotted points.

Statistic over all 160 points	Measured	Expected if unbiased
points with $\lvert z\rvert < 1$	71.2%	68.3%
points with $\lvert z\rvert < 2$	95.0%	95.4%
points with $\lvert z\rvert < 3$	99.4%	99.7%
largest $\lvert z\rvert$	3.02	reached by chance in 33% of runs
mean of z (bias)	+0.079	0 ± 0.08
mean of z^2	0.97	1 ± 0.11

Conclusion. Every simulated curve agrees with its exact theory, and the 160 deviations are spread exactly like Monte Carlo noise should be. The share of points within 1σ, 2σ and 3σ matches the normal distribution, the mean of z shows no bias, and the mean of z^2 is 1 within its accuracy. The largest deviation, 3.02σ (BPSK in fading at 6 dB, +1.3%), is the only point beyond 3σ. For 160 independent points, a maximum this large happens by chance in about one run in three. The only *systematic* difference in this report is between the 4-QAM fading simulation and the *approximate* formula of the assignment table, explained in Sec. 2.3.1.

2 Problem 1 — SER in AWGN and Rayleigh fading

Problem 1 measures the SER of BPSK and 4-QAM over the channel

$$y = hx + w, \qquad x \in \mathcal{X}, \qquad w \sim \mathcal{CN}(0, 1/\mathrm{SNR}), \tag{4}$$

with $h = 1$ in Task 1 (AWGN) and $h \sim \mathcal{CN}(0, 1)$ in Task 2 (Rayleigh fading). Both tasks use the unit-energy constellations, the ML detector and the error counting of Sec. 1. In both channels the receiver knows h, so the ML decision is

$$\hat{\imath} = \arg\min_i |y - h\,u_i|^2. \tag{5}$$

Each task below first derives the theoretical SER, then describes the simulation and shows its figure. The results are discussed in Sec. 2.3.

2.1 Task 1: AWGN channel

Theoretical expressions. With $h = 1$ the receiver sees $y = x + w$. The real and imaginary parts of w are independent, each $\mathcal{N}(0, 1/(2\mathrm{SNR}))$. The Gaussian tail function $Q(a) = \frac{1}{2}\,\mathrm{erfc}(a/\sqrt{2})$ is implemented as its own function (`qfunc`).

- **BPSK.** The points ± 1 are separated by the threshold $\mathrm{Re}\{y\} = 0$, at distance 1 from each point. An error happens when the real part of the noise is larger than 1 in the wrong direction:

$$P_{\mathrm{e}}^{\mathrm{BPSK}} = \Pr\left[\mathcal{N}(0, \tfrac{1}{2\mathrm{SNR}}) > 1\right] = Q(\sqrt{2\mathrm{SNR}}). \tag{6}$$

- **4-QAM.** The points $(\pm 1 \pm j)/\sqrt{2}$ are two BPSK signals, one on I and one on Q, each with amplitude $1/\sqrt{2}$ (half the energy) and its own independent noise. One component is wrong with probability $q = Q((1/\sqrt{2})/\sqrt{1/(2\mathrm{SNR})}) = Q(\sqrt{\mathrm{SNR}})$. The symbol is correct only if both components are correct:

$$P_{\mathrm{e}}^{\mathrm{4QAM}} = 1 - (1-q)^2 = 2Q(\sqrt{\mathrm{SNR}}) - Q^2(\sqrt{\mathrm{SNR}}). \tag{7}$$

  An error in one or in both components counts as one symbol error.

Simulation. The script `problem1/awgn_ser.py` runs Algorithm 1 for $\mathrm{SNR}_{\mathrm{dB}} \in \{0, 1, \ldots, 12\}$. For each modulation and SNR it draws $N = 10^6$ equiprobable indices, sends $x = u_i$ through $y = x + w$, detects with (5) and counts the errors. The expressions (6) and (7) are evaluated at the same SNR points. Fig. 1 shows both schemes on one figure: simulation as markers, theory as lines.

2.2 Task 2: Rayleigh fading channel

Channel. A new $h \sim \mathcal{CN}(0, 1)$ is drawn for every symbol and is known at the receiver (coherent detection). For a given h, the link is an AWGN link whose SNR is multiplied by the channel power gain $g = |h|^2$. Since $h \sim \mathcal{CN}(0, 1)$, g is exponential, $f_g(g) = e^{-g}$ for $g \geq 0$, with $\mathbb{E}[g] = 1$. So SNR is also the average received SNR.

Theoretical expressions. Each fading result is the AWGN result for a fixed g, averaged over the distribution of g.

- **BPSK.** For a fixed g, (6) becomes $Q(\sqrt{2g\mathrm{SNR}})$. Averaging over g gives the result of [1, Sec. 5.2]:

$$P_{\mathrm{e}}^{\mathrm{BPSK}} = \int_0^\infty Q(\sqrt{2g\mathrm{SNR}})\, e^{-g}\, dg = \frac{1}{2}\Big(1 - \sqrt{\frac{\mathrm{SNR}}{1+\mathrm{SNR}}}\Big). \tag{8}$$

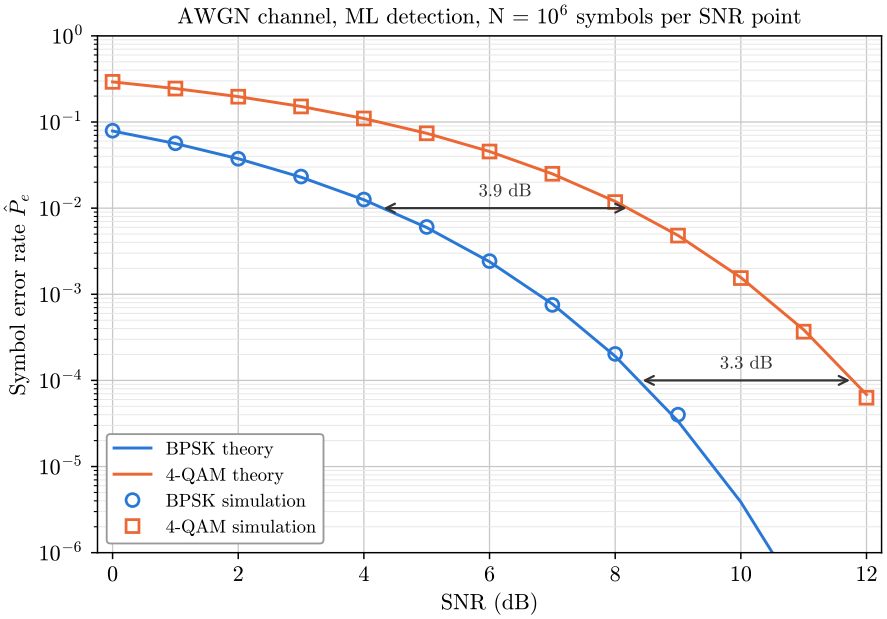


Figure 1: Problem 1, Task 1: SER of BPSK and 4-QAM in the AWGN channel ($h = 1$) with ML detection. Markers: Monte Carlo simulation with $N = 10^6$ symbols per SNR point, $\mathrm{SNR_{dB}} \in \{0, 1, \ldots, 12\}$; only points with $n_{\mathrm{err}} \geq 10$ are shown. Lines: theory (6) and (7). Arrows: the measured SNR gap between the two schemes, 3.9 dB at 10^{-2} and 3.3 dB at 10^{-4}. **The BPSK–4-QAM gap is not constant: it shrinks towards 3 dB as the SNR grows** (Sec. 2.3.3).

- **4-QAM, formula of the assignment table.** Twice the BPSK result at half the SNR:

$$P_{\mathrm{e}}^{4\mathrm{QAM}} \approx 2 \cdot \frac{1}{2}\Big(1 - \sqrt{\frac{\mathrm{SNR}/2}{1 + \mathrm{SNR}/2}}\Big). \tag{9}$$

  This is the probability of an I error plus the probability of a Q error. It leaves out the event that both are wrong, so it is an approximation (an upper bound).
- **4-QAM, exact (extra, for checking).** Averaging the complete AWGN expression (7), with $q = Q(\sqrt{g\mathrm{SNR}})$, over g gives, with Craig's formula (Sec. 1.4),

$$P_{\mathrm{e}}^{4\mathrm{QAM}} = \int_0^\infty \left[2q - q^2\right] e^{-g}\, dg = (1 - \mu) - \tfrac{1}{4}\Big[1 - \tfrac{4}{\pi}\,\mu \arctan\tfrac{1}{\mu}\Big], \qquad \mu = \sqrt{\frac{\mathrm{SNR}/2}{1 + \mathrm{SNR}/2}}. \tag{10}$$

  The first term, $1 - \mu$, is exactly the table formula (9); the second is the dropped "both wrong" term. This curve is only used to explain the simulated 4-QAM points; the comparison asked for in Task 2 uses (9).

Simulation. The script `problem1/rayleigh_ser.py` runs Algorithm 1 for $\mathrm{SNR_{dB}} \in \{0, 2, \ldots, 40\}$, with a fresh h and w for every symbol and $N = 10^6$ symbols per modulation and SNR point. The detector (5) uses the known h. Fig. 2 shows the simulation with the expressions (8)–(10). Fig. 3 is an extra figure that places the AWGN theory of Task 1 next to the fading results; it is used in the discussion of Sec. 2.3.2.

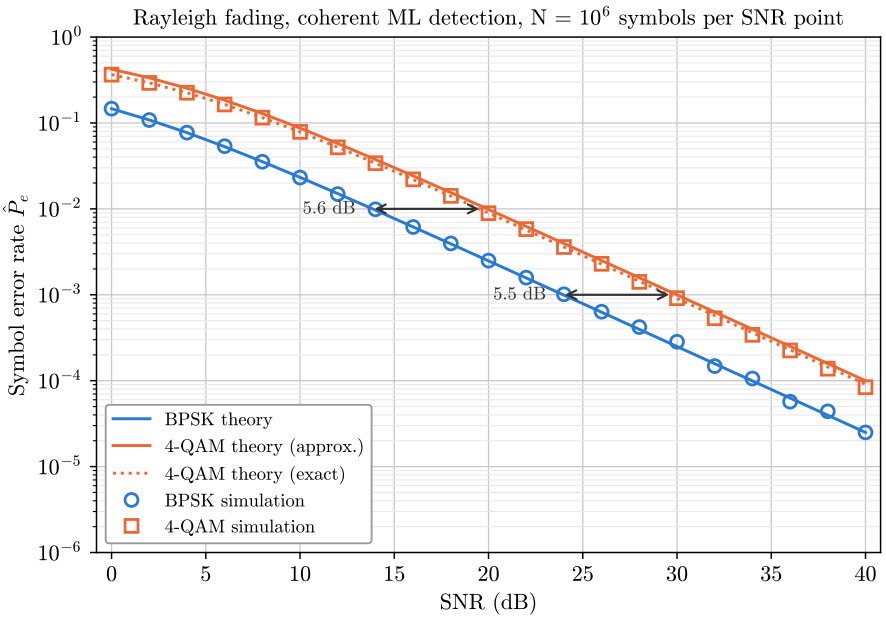


Figure 2: Problem 1, Task 2: SER of BPSK and 4-QAM in Rayleigh fading with coherent ML detection. Markers: simulation with $N = 10^6$ symbols per SNR point, $\mathrm{SNR}_{\mathrm{dB}} \in \{0, 2, \ldots, 40\}$, points with $n_{\mathrm{err}} \geq 10$. Solid lines: assignment table (8) and (9) (an approximation for 4-QAM). Dotted: exact 4-QAM SER (10). Arrows: the measured BPSK–4-QAM gap, 5.6 dB at 10^{-2} and 5.5 dB at 10^{-3}. **The gap is nearly constant, about 5.5 dB: both curves are parallel lines of slope −1** (Sec. 2.3.3).

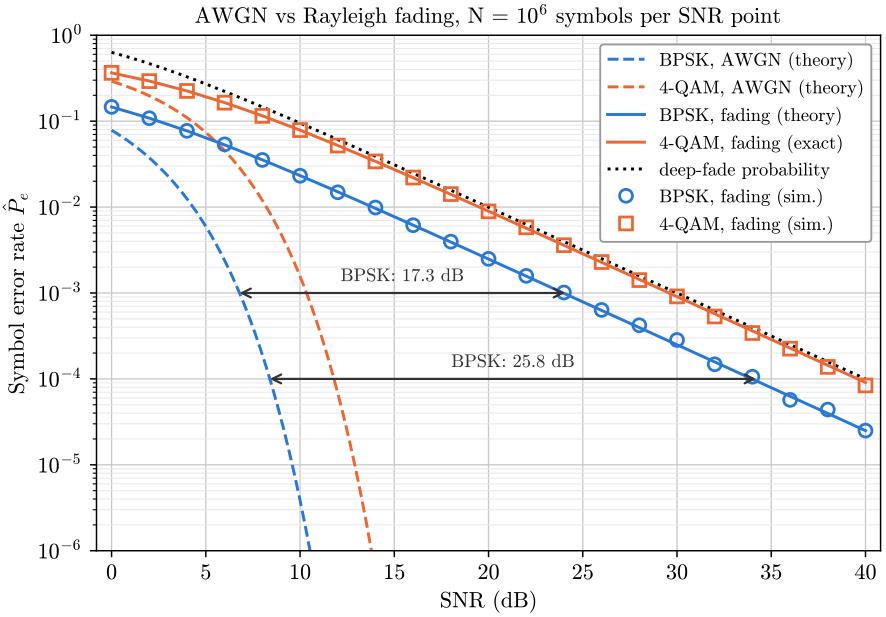


Figure 3: Extra figure: AWGN (dashed, theory) against Rayleigh fading (solid lines: theory; markers: the simulation of Fig. 2, $N = 10^6$ symbols per SNR point, 0–40 dB). Dotted: deep-fade probability $\Pr[|h|^2\mathrm{SNR} < 1] = 1 - e^{-1/\mathrm{SNR}} \approx 1/\mathrm{SNR}$. Arrows: the extra SNR that BPSK needs in fading, 17.3 dB at 10^{-3} and 25.8 dB at 10^{-4}. **The AWGN–fading gap is not constant: it grows as the target SER falls** (Sec. 2.3.3).

2.3 Discussion

The assignment asks for two points: the agreement between simulation and theory, with the reason for any deviation (Sec. 2.3.1), and the difference between the AWGN and fading curves at high SNR, explained with the deep-fade argument (Sec. 2.3.2). Sec. 2.3.3 explains the gaps marked by the arrows in Figs. 1–3.

2.3.1 Agreement between simulation and theory

- **AWGN (Fig. 1, Table 7).** Both schemes match theory: max $|z|$ is 2.59 (BPSK) and 1.95 (4-QAM). The largest *relative* deviations occur where few errors are seen. For example, BPSK at 9 dB has 40 errors where 33.6 are expected (+19%, only 1.1σ). At 10, 11 and 12 dB, BPSK gives 6, 0 and 0 errors (3.9, 0.26 and 0.01 expected). These points are not plotted (G.7): 10^6 symbols cannot measure such small error rates.
- **Fading, BPSK.** The simulation follows the exact formula $\frac{1}{2}(1 - \sqrt{\mathrm{SNR}/(1+\mathrm{SNR})})$, within $\pm 2.5\%$ wherever $n_{\mathrm{err}} > 1000$. With fewer errors the scatter grows, as the table of Sec. 1.3 predicts. The largest relative deviation is +14% at 30 dB (284 errors against 250 expected, 2.2σ).
- **Fading, 4-QAM: a systematic deviation from the table formula.** The simulated SER is *below* the table formula at every SNR: 13.5% lower at 0 dB and 7–11% lower from 10 to 30 dB. That is up to 116σ, so it is not noise. The simulation is not wrong: it matches the *exact* curve (dotted in Fig. 2, max $|z| = 1.78$). The reasons:
  - Per symbol, the error probability is $2q - q^2$ with $q = Q(\sqrt{|h|^2\mathrm{SNR}})$ the error probability of one component. The table formula keeps only $\mathbb{E}[2q]$ and drops $-\mathbb{E}[q^2]$, the event that I *and* Q are both wrong. It is therefore an upper bound.
  - In AWGN, $q^2 \ll q$ at high SNR, so the dropped term does not matter. In fading it does: errors happen mostly in deep fades, and there both components see the same small $|h|$ and fail together.
  - At high SNR, $\mathbb{E}[2q] \to 1/\mathrm{SNR}$ and $\mathbb{E}[q^2] \to \frac{1}{4}(1 - \frac{2}{\pi})/\mathrm{SNR} = 0.091/\mathrm{SNR}$. The exact SER is therefore $0.909/\mathrm{SNR}$, and the table formula is 10% too high. The simulation output confirms this: the formula exceeds the exact value by 15.8% at 0 dB and by 10.0% from 30 dB on.
  - On the dB axis this is only $10\log_{10}(1/0.909) = 0.41$ dB, which is why the two lines look so close in Fig. 2.

2.3.2 AWGN versus fading at high SNR: the deep-fade argument

What the figures show.

- *AWGN:* the SER falls *exponentially* with SNR, because $Q(x) \le \frac{1}{2}e^{-x^2/2}$. The curves are a steep "waterfall": BPSK goes from 10^{-3} at 6.8 dB to 10^{-4} at 8.4 dB, one decade in only 1.6 dB.
- *Rayleigh fading:* the curves become straight lines of slope -1, so $P_{\mathrm{e}} \propto \mathrm{SNR}^{-1}$: one decade per 10 dB. The fitted slope over 30–40 dB is 1.02 for both schemes. At high SNR, $P_{\mathrm{e}}^{\mathrm{BPSK}} \approx 1/(4\mathrm{SNR})$ and $P_{\mathrm{e}}^{4\mathrm{QAM}} \approx 0.909/\mathrm{SNR}$.

Why: errors come from deep fades [1, Sec. 5.4].

1. For a given channel, the link is an AWGN link whose SNR is scaled by $|h|^2$: $P_{\mathrm{e}}(h) = Q(\sqrt{2|h|^2\mathrm{SNR}})$ for BPSK.
2. This gives two regimes:
   - if $|h|^2\mathrm{SNR} \gg 1$, $P_{\mathrm{e}}(h)$ is exponentially small, so good channels contribute almost no

errors;

- if $|h|^2\text{SNR} \lesssim 1$ (a *deep fade*), $P_\text{e}(h)$ is of order one (at least $Q(\sqrt{2}) = 0.079$).

3. Because $|h|^2 \sim \text{Exp}(1)$, a deep fade is not rare:

$$\Pr\left[|h|^2 < 1/\text{SNR}\right] = 1 - e^{-1/\text{SNR}} \approx \frac{1}{\text{SNR}}.$$

   This is the dotted line in Fig. 3. It runs right along the fading curves, with the same slope.

4. Hence $P_\text{e} \approx \Pr[\text{deep fade}] \times O(1) \propto 1/\text{SNR}$. More transmit power makes deep fades only proportionally rarer, so the SER also falls only linearly.

Simulation evidence. The Problem 1 script also records $|h|^2\text{SNR}$ for every BPSK symbol and checks which errors happen during a deep fade (`results/q1/rayleigh_summary.txt`):

SNR_dB	10	20	30	40
symbols sent in a deep fade	9.5%	1.0%	0.10%	0.010%
errors (out of 10^6 symbols)	23 570	2 506	236	24
errors that happened in a deep fade	77%	74%	70%	83%

Only about 1/SNR of the symbols are sent in a deep fade, yet these few symbols cause about three quarters of all errors at every SNR.

The price of fading. SNR needed for a target SER:

Scheme	Target P_e	AWGN (theory)	Rayleigh (simulation)	Extra SNR needed
BPSK	10^{-3}	6.8 dB	24.1 dB	17.3 dB
BPSK	10^{-4}	8.4 dB	34.2 dB	25.8 dB
4-QAM	10^{-3}	10.4 dB	29.6 dB	19.2 dB
4-QAM	10^{-4}	11.8 dB	39.3 dB	27.5 dB

The penalty grows as the target SER falls (arrows in Fig. 3). Reducing it is the purpose of the diversity schemes of Problems 2 and 3.

2.3.3 Gaps between the curves (arrows in the figures)

- **BPSK versus 4-QAM in AWGN: not constant** (Fig. 1). Measured: 3.87 dB at 10^{-2}, 3.54 dB at 10^{-3} and 3.30 dB at 10^{-4} (theory 3.89, 3.56 and 3.40 dB).
  - At equal symbol energy, each 4-QAM component has half the energy of a BPSK symbol: $Q(\sqrt{\text{SNR}})$ against $Q(\sqrt{2\text{SNR}})$. That alone is a shift of exactly 3 dB.
  - 4-QAM also has the factor 2 in front of $Q(\cdot)$ (either component can fail). On a steep waterfall this factor costs only a small extra SNR, which shrinks as the SNR grows, so the gap tends to 3.01 dB.
- **BPSK versus 4-QAM in fading: constant, about 5.6 dB** (Fig. 2). Measured: 5.56 dB at 10^{-2} and 5.53 dB at 10^{-3} (exact theory 5.66 and 5.61 dB). At 10^{-4} the measured value, 5.11 dB, is less accurate: there the 4-QAM curve is read between its last two points, with only 138 and 84 errors.
  - Both curves have slope -1, so at high SNR they are parallel, $P_\text{e} \approx c/\text{SNR}$, and only the constant c depends on the modulation: $c = 0.25$ for BPSK and 0.909 for 4-QAM. The offset is $10\log_{10}(0.909/0.25) = 5.61$ dB.
  - The offset is larger than in AWGN because on a slope-1 line a factor in front of the SER is never small. Half the energy per component doubles c (3.01 dB), two components double it again (3.01 dB), and the joint-error term removes 9% (-0.41 dB).
- **AWGN versus fading: not constant, it grows** (Fig. 3). For BPSK it is 17.3 dB at 10^{-3} and 25.8 dB at 10^{-4}. One curve falls exponentially and the other as 1/SNR, so they cannot be parallel.

3 Problem 2 — Receive diversity (SIMO $1 \times L$), 4-QAM

Model. $\mathbf{y} = \mathbf{h}x + \mathbf{w}$ with $h_\ell \sim \mathcal{CN}(0,1)$ i.i.d., $w_\ell \sim \mathcal{CN}(0, 1/\mathrm{SNR})$ i.i.d., and a new $\mathbf{h}$ for every symbol, known at the receiver. SNR is the average SNR *per receive antenna*. For every L and SNR, MRC and SC process the *same* symbols, channels and noise, so their comparison is as fair as possible.

3.1 Task 1: decision statistics as equivalent scalar channels

Receiver	Decision statistic r	Gain g	Noise variance	Output SNR
MRC (as implemented)	$\mathbf{h}^H\mathbf{y}/\lVert\mathbf{h}\rVert$	$\lVert\mathbf{h}\rVert$	$1/\mathrm{SNR}$	$\lVert\mathbf{h}\rVert^2\mathrm{SNR}$
MRC (not normalised)	$\mathbf{h}^H\mathbf{y}$	$\lVert\mathbf{h}\rVert^2$	$\lVert\mathbf{h}\rVert^2/\mathrm{SNR}$	$\lVert\mathbf{h}\rVert^2\mathrm{SNR}$
SC	$y_{\ell^\star}$, $\ell^\star = \arg\max_\ell \lvert h_\ell\rvert^2$	$h_{\ell^\star}$	$1/\mathrm{SNR}$	$\max_\ell \lvert h_\ell\rvert^2\mathrm{SNR}$

In the code these are `mrc_combine` and `selection_combine` (`common/combining.py`).

MRC [2, Secs. 3.1–3.2]. Each branch is weighted by its conjugate channel gain and the branches are added. The code normalises the sum by $\lVert\mathbf{h}\rVert$:

$$r = \frac{\mathbf{h}^H\mathbf{y}}{\lVert\mathbf{h}\rVert} = \lVert\mathbf{h}\rVert\, x + \underbrace{\frac{\mathbf{h}^H\mathbf{w}}{\lVert\mathbf{h}\rVert}}_{n}, \qquad \mathrm{Var}(n) = \frac{\sum_\ell \lvert h_\ell\rvert^2 \cdot (1/\mathrm{SNR})}{\lVert\mathbf{h}\rVert^2} = \frac{1}{\mathrm{SNR}}.$$

This is (1) with the real, positive gain $g = \lVert\mathbf{h}\rVert$, and the Problem 1 detector $\hat{\imath} = \arg\min_i \lvert r - \lVert\mathbf{h}\rVert u_i\rvert^2$ is applied to it.

Why this is exactly the ML rule. With L antennas the ML rule is $\arg\min_i \lVert\mathbf{y} - \mathbf{h}u_i\rVert^2$. Expanding and using $\mathbf{h}^H\mathbf{y} = \lVert\mathbf{h}\rVert\, r$:

$$\lVert\mathbf{y} - \mathbf{h}u_i\rVert^2 = \lVert\mathbf{y}\rVert^2 - 2\lVert\mathbf{h}\rVert\,\mathrm{Re}(u_i^* r) + \lVert\mathbf{h}\rVert^2\lvert u_i\rvert^2 = \lvert r - \lVert\mathbf{h}\rVert u_i\rvert^2 + \underbrace{\left(\lVert\mathbf{y}\rVert^2 - \lvert r\rvert^2\right)}_{\text{same for every } i}.$$

The last term does not depend on i, so both rules pick the same symbol. Without the normalisation, the statistic is $\mathbf{h}^H\mathbf{y} = \lVert\mathbf{h}\rVert^2 x + \mathbf{h}^H\mathbf{w}$, with gain $\lVert\mathbf{h}\rVert^2$ and noise variance $\lVert\mathbf{h}\rVert^2/\mathrm{SNR}$. Its metric is $\lvert\mathbf{h}^H\mathbf{y} - \lVert\mathbf{h}\rVert^2 u_i\rvert^2 = \lVert\mathbf{h}\rVert^2\,\lvert r - \lVert\mathbf{h}\rVert u_i\rvert^2$, the same metric multiplied by a positive constant, so it is also ML.

SC [2, Sec. 3.5]. The receiver keeps only the strongest antenna, $r = y_{\ell^\star} = h_{\ell^\star}x + w_{\ell^\star}$, and detects it with the coherent Problem 1 detector ($g = h_{\ell^\star}$). The choice of $\ell^\star$ depends only on $\mathbf{h}$, which is independent of $\mathbf{w}$, so $w_{\ell^\star}$ is still $\mathcal{CN}(0, 1/\mathrm{SNR})$.

Numerical check (`results/q2/summary.txt`). At 10 dB with 10^6 symbols, the measured noise variance of the decision statistic, divided by the predicted $1/\mathrm{SNR}$, is 0.9986 (MRC) and 0.9987 (SC) for $L = 2$, and 0.9993 (MRC) and 1.0003 (SC) for $L = 4$. Each is within 0.15% of 1, the Monte Carlo accuracy of 10^6 samples. The mean gains $\mathbb{E}\lVert\mathbf{h}\rVert^2 = L$ and $\mathbb{E}\max_\ell \lvert h_\ell\rvert^2 = \sum_{k=1}^{L} 1/k$ are also reproduced to 0.01 dB.

3.2 Task 2: simulation

The SER of MRC and SC was estimated for $L \in \{1, 2, 4\}$ and $\mathrm{SNR}_{\mathrm{dB}} \in \{0, 2, \ldots, 24\}$ with $N = 10^6$ symbols per point. All curves are plotted on one figure (Fig. 4): one colour per L, solid lines for MRC and dashed lines for SC. Fig. 5 shows the same points against the exact theory of Sec. 1.4.

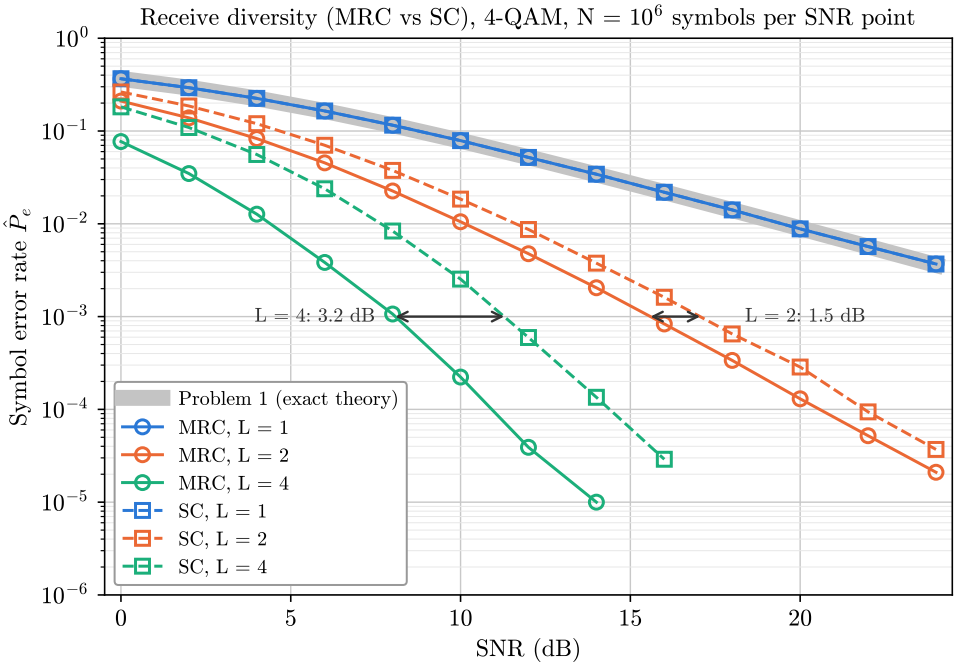


Figure 4: Problem 2: SER of 4-QAM with $L \in \{1, 2, 4\}$ receive antennas. MRC: solid lines, circles; SC: dashed lines, squares; one colour per L. Grey band: exact single-antenna 4-QAM SER of Problem 1. Monte Carlo with $N = 10^6$ symbols per SNR point, $\mathrm{SNR}_{\mathrm{dB}} \in \{0, 2, \ldots, 24\}$ (average SNR per antenna); points with $n_{\mathrm{err}} < 10$ are omitted, so the $L = 4$ curves stop at 14 and 16 dB. Arrows: the MRC–SC gap at 10^{-3}. For $L = 1$ the two curves lie exactly on top of each other.

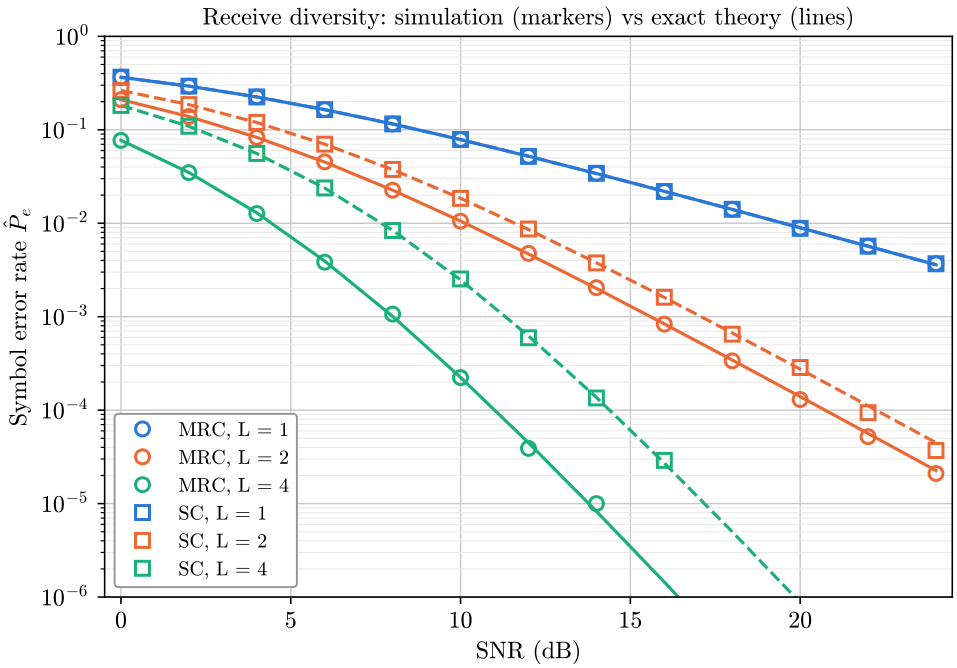


Figure 5: Extra figure: the simulated points of Fig. 4 ($N = 10^6$ symbols per SNR point, 0–24 dB) against the exact theory of Sec. 1.4 (lines; MRC solid, SC dashed).

3.3 Discussion

Decision statistics. These are derived in Sec. 3.1. Both receivers give $r = g\,x + n$ with $n \sim \mathcal{CN}(0, 1/\mathrm{SNR})$, so the Problem 1 detector is reused unchanged.

Consistency check ($L = 1$).

- With one antenna, MRC is only a phase rotation, $r = (h^*/|h|)\,y$, and SC has only one branch to choose. Both reduce to coherent detection on $y = hx + w$. They also process the same realisations, so their error counts are *identical* at every SNR (checked in the code), and the two curves coincide in Fig. 4.
- The $L = 1$ curve lies on the 4-QAM fading curve of Problem 1. It agrees with the exact theory within 1.85σ, and with an independent re-run of the Problem 1 simulation within 1.69σ.
- All MRC and SC curves agree with exact theory (Fig. 5, Table 9); every plotted point is within 2.01σ.

High-SNR slope versus diversity order. MRC and SC both have diversity order L, i.e. $P_\mathrm{e} \propto \mathrm{SNR}^{-L}$. An error now needs *all* L independent branches to fade at once. With $\epsilon = 1/\mathrm{SNR}$:

$$\Pr\left[\|\mathbf{h}\|^2 < \epsilon\right] \approx \frac{\epsilon^L}{L!} \quad \text{(MRC)}, \qquad \Pr\left[\max_\ell |h_\ell|^2 < \epsilon\right] = \left(1 - e^{-\epsilon}\right)^L \approx \epsilon^L \quad \text{(SC)}. \tag{11}$$

Table 3 gives the slope $-\Delta \log_{10} \hat{P}_\mathrm{e} / \Delta \log_{10} \mathrm{SNR}$ at the high-SNR end of each curve.

Table 3: High-SNR slopes over the last four points with at least 50 errors: local slope of each 2 dB step, least-squares fit through the four points, the exact theory fitted over the same points, and the exact theory at 60–70 dB.

			Fitted slope		Exact theory,	Diversity
Receiver	Fit range	Local slopes	simulation	exact theory	60–70 dB	order
MRC / SC, $L = 1$	18–24 dB	1.02, 0.95, 0.94	0.97	0.99	1.00	1
MRC, $L = 2$	16–22 dB	1.97, 2.07, 1.99	2.01	1.96	2.00	2
SC, $L = 2$	16–22 dB	1.98, 1.78, 2.42	2.03	1.93	2.00	2
MRC, $L = 4$	4–10 dB	2.60, 2.78, 3.40	2.91	2.94	4.00	4
SC, $L = 4$	8–14 dB	2.59, 3.15, 3.22	3.00	2.98	4.00	4

- **$L = 1$ and $L = 2$:** the fitted slopes (0.97, 2.01 and 2.03) are the predicted diversity orders 1 and 2. A single 2 dB step is less accurate (about ±0.2–0.3 with a few hundred errors per point), which explains the 1.78 and 2.42 in the SC row.
- **$L = 4$:** the measured slope is about 3, not 4. This is not a contradiction: L is the *asymptotic* slope. With 10^6 symbols the $L = 4$ curves can only be measured down to about 10^{-5} (up to 10–14 dB), where they are still bending. Exact theory over the same points gives the same slopes (2.94 and 2.98). It reaches 4.00 only at 60–70 dB, where P_e is between 10^{-22} and 10^{-28}, far below anything a simulation can reach. The slope does grow from 2.6 to 3.4 along the measured part, as expected.

SNR gap between MRC and SC (arrows in Fig. 4). Table 4 gives the SNR each receiver needs for a target SER.

- **The gap is almost constant for each L**: about 1.5 dB for $L = 2$ and 3.2–3.5 dB for $L = 4$. MRC and SC have the same slope L, so their curves are parallel at high SNR. (The 1.32 dB at 10^{-4} for $L = 2$ is read between points with only 52–286 errors, so it is accurate to only about ±0.2 dB.)
- **Origin: array gain** [2, Sec. 3.4], the gain in *average* output SNR over one antenna, $\mathbb{E}[\mathrm{SNR}_\mathrm{out}]/\mathrm{SNR}$:

Table 4: SNR (dB) needed for a target SER, read off the simulated curves (exact theory in brackets), and the gap SC − MRC.

		$P_e = 10^{-2}$	$P_e = 10^{-3}$	$P_e = 10^{-4}$
$L = 2$	MRC	10.13 (10.14)	15.59 (15.60)	20.57 (20.73)
	SC	11.63 (11.61)	17.05 (17.09)	21.89 (22.24)
	gap	**1.50** (1.47)	**1.46** (1.49)	**1.32** (1.50)
$L = 4$	MRC	4.40 (4.46)	8.08 (8.01)	10.92 (11.02)
	SC	7.66 (7.69)	11.28 (11.34)	14.39 (14.40)
	gap	**3.26** (3.23)	**3.20** (3.33)	**3.47** (3.38)

  - MRC adds the signals of all L antennas coherently, while the noises add incoherently: $\mathbb{E}\|\mathbf{h}\|^2 = L$, i.e. 3.01 dB ($L = 2$) and 6.02 dB ($L = 4$).
  - SC throws away the energy received on the other $L-1$ antennas: $\mathbb{E}\max_\ell |h_\ell|^2 = \sum_{k=1}^{L} 1/k$, i.e. only 1.76 dB ($L = 2$) and 3.19 dB ($L = 4$).
  - The difference, 1.25 dB ($L = 2$) and 2.83 dB ($L = 4$), is most of the gap. It also explains why the gap grows with L: MRC's array gain grows like L, SC's only like $\ln L$.
- **Why the gap is a little larger than 1.25 / 2.83 dB.** At high SNR the SER depends on how often the output SNR is *small*, not on its mean. By (11), SC is in a deep fade $L!$ times as often as MRC, so its SER is $L!$ times higher. On curves of slope L that is a horizontal shift of $10\log_{10}(L!)/L$: 1.51 dB ($L = 2$) and 3.45 dB ($L = 4$). The measured gaps lie between the array-gain value and this limit.
- **Between different L, the gap is not constant.** The slopes differ (1, 2 and 4), so the curves for different L spread apart as the SER falls. This growing gap is the diversity gain.

4 Problem 3 — Transmit diversity (MISO 2×1), 4-QAM

Model. $h_1, h_2 \sim \mathcal{CN}(0,1)$ i.i.d., known at the receiver, and $w \sim \mathcal{CN}(0, 1/\mathrm{SNR})$. For a fair comparison, the total energy transmitted per channel use, summed over both antennas, is 1, as for a single antenna. So SNR is the total transmit energy divided by the noise variance.

4.1 Task 1: transmission schemes

Single antenna (reference). $s = x$, $y = hx + w$, with coherent ML detection: the fading channel of Problem 1.

Transmit beamforming [2, Sec. 4.1] (channel known at the transmitter). Antenna ℓ sends $v_\ell x$ with the weights

$$\mathbf{v} = \frac{\mathbf{h}^*}{\|\mathbf{h}\|} = \frac{1}{\sqrt{|h_1|^2 + |h_2|^2}} \begin{pmatrix} h_1^* \\ h_2^* \end{pmatrix}, \qquad y = h_1 v_1 x + h_2 v_2 x + w = \|\mathbf{h}\|\, x + w.$$

- Each path arrives with the real, positive gain $h_\ell v_\ell = |h_\ell|^2/\|\mathbf{h}\|$, so the two paths add in phase.
- Equivalent scalar channel: $r = y$, gain $\|\mathbf{h}\|$, noise variance $1/\mathrm{SNR}$, output SNR $\|\mathbf{h}\|^2\mathrm{SNR}$. This is the same as MRC with $L = 2$. Detection: $\arg\min_i |y - \|\mathbf{h}\| u_i|^2$.
- *Scaling:* $\|\mathbf{v}\| = 1$, so the energy per channel use is $|v_1 x|^2 + |v_2 x|^2 = |x|^2$, with mean 1. No further scaling is needed.

Alamouti scheme [2, Sec. 4.2] (no channel knowledge at the transmitter). Two symbols are sent in

two channel uses. The transmission matrix, with rows for channel uses and columns for antennas, is

$$\mathbf{S} = \frac{1}{\sqrt{2}} \begin{pmatrix} x_1 & x_2 \\ -x_2^* & x_1^* \end{pmatrix}, \qquad \begin{aligned} y_1 &= \tfrac{1}{\sqrt{2}} (h_1 x_1 + h_2 x_2) + w_1, \\ y_2 &= \tfrac{1}{\sqrt{2}} (-h_1 x_2^* + h_2 x_1^*) + w_2. \end{aligned} \tag{12}$$

The channel is constant over the two channel uses of a block and independent across blocks. Conjugating the second sample gives a linear model whose channel matrix has orthogonal columns:

$$\begin{pmatrix} y_1 \\ y_2^* \end{pmatrix} = \frac{1}{\sqrt{2}} \underbrace{\begin{pmatrix} h_1 & h_2 \\ h_2^* & -h_1^* \end{pmatrix}}_{\mathbf{H}} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} w_1 \\ w_2^* \end{pmatrix}, \qquad \mathbf{H}^H \mathbf{H} = \|\mathbf{h}\|^2 \, \mathbf{I}.$$

Alamouti combining projects onto the two columns (normalised by $\|\mathbf{h}\|$):

$$\begin{aligned} r_1 &= \frac{h_1^* y_1 + h_2 y_2^*}{\|\mathbf{h}\|} = \frac{\|\mathbf{h}\|}{\sqrt{2}} \, x_1 + n_1, \\ r_2 &= \frac{h_2^* y_1 - h_1 y_2^*}{\|\mathbf{h}\|} = \frac{\|\mathbf{h}\|}{\sqrt{2}} \, x_2 + n_2. \end{aligned} \tag{13}$$

- The other symbol cancels exactly. The noises $n_1 = (h_1^* w_1 + h_2 w_2^*)/\|\mathbf{h}\|$ and $n_2 = (h_2^* w_1 - h_1 w_2^*)/\|\mathbf{h}\|$ are independent $\mathcal{CN}(0, 1/\mathrm{SNR})$ (their covariance is $\mathbf{H}^H\mathbf{H}/(\|\mathbf{h}\|^2\mathrm{SNR}) = \mathbf{I}/\mathrm{SNR}$), so detecting x_1 and x_2 separately is ML.
- Each r_k is a scalar channel with gain $\|\mathbf{h}\|/\sqrt{2}$, noise variance $1/\mathrm{SNR}$ and output SNR $\|\mathbf{h}\|^2\mathrm{SNR}/2$. It is detected by the Problem 1 detector with $g = \|\mathbf{h}\|/\sqrt{2}$.
- *Scaling:* every entry of $\mathbf{S}$ carries $1/\sqrt{2}$, so a channel use carries $(|x_1|^2 + |x_2|^2)/2$, with mean 1, and each antenna carries half of it.
- The rate is one symbol per channel use. $N/2 = 5 \times 10^5$ blocks are simulated per SNR point, and the SER is counted over all $N = 10^6$ symbols.

Numerical check of the energy constraint. The simulation averages the energy actually sent, $\sum_\ell |s_\ell|^2$, over all channel uses at every SNR point (`results/q3/summary.txt`):

Scheme	Scaling	Total energy (min / max over SNR points)	Per antenna
Single antenna	$s = x$	1.000000 / 1.000000	1.0000
Beamforming	$s_\ell = (h_\ell^*/\|\mathbf{h}\|)\, x$	1.000000 / 1.000000	0.5000 / 0.5000
Alamouti	entries of $\mathbf{S}$, $\pm x_k^{(*)}/\sqrt{2}$	1.000000 / 1.000000	0.5000 / 0.5000

All schemes, and MRC (one transmit antenna sending x), send exactly the energy of the single-antenna system. For 4-QAM this holds in every single channel use, not just on average, because $|x|^2 = 1$ for every point.

4.2 Task 2: simulation

Every curve agrees with its exact theory (Fig. 7, Table 10); the largest deviation is 2.55σ. Beamforming and the independent MRC simulation agree within 1.51σ.[1]

4.3 Discussion

Diversity order (Table 5).

[1]The single-antenna curve is identical to the $L = 1$ curve of Problem 2. Both scripts start from seed 2026 and draw the symbols, channels and noise in the same order.

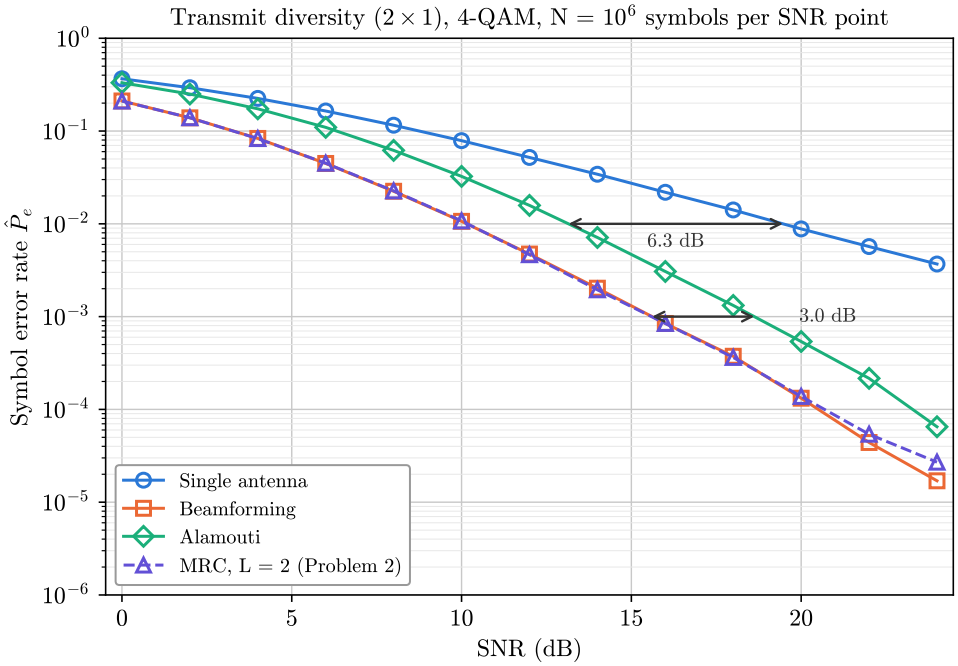


Figure 6: Problem 3: SER of 4-QAM with a single antenna, transmit beamforming and the Alamouti scheme (2×1), and MRC with $L = 2$ from Problem 2. All schemes transmit a total energy of 1 per channel use. Monte Carlo with $N = 10^6$ symbols per SNR point (Alamouti: 5×10^5 blocks of two symbols), $\mathrm{SNR}_{\mathrm{dB}} \in \{0, 2, \ldots, 24\}$; points with $n_{\mathrm{err}} < 10$ are omitted. Arrows: Alamouti is 3.0 dB behind beamforming at 10^{-3}, and the single antenna is 6.3 dB behind Alamouti at 10^{-2}, a gap that grows. The beamforming and MRC curves lie on top of each other.

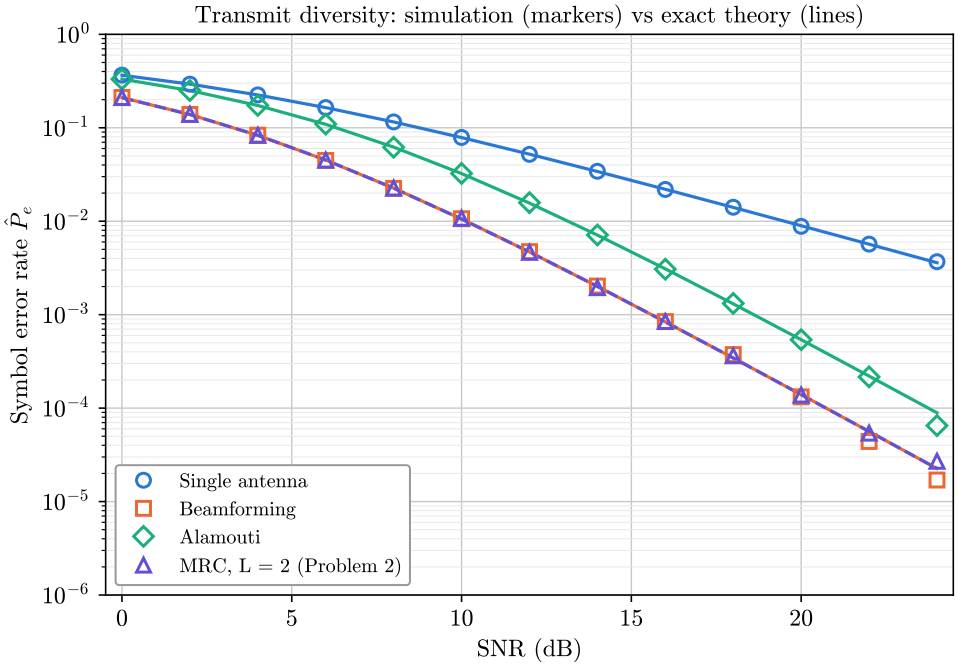


Figure 7: Extra figure: the simulated points of Fig. 6 ($N = 10^6$ symbols per SNR point, 0–24 dB) against the exact theory of Sec. 1.4 (lines). Beamforming and MRC have the same theoretical curve.

Table 5: High-SNR slopes over the last four points with at least 50 errors, as in Table 3.

Scheme	Fit range	Local slopes	Fitted slope		Exact theory,	Diversity
			simulation	exact theory	60–70 dB	order
Single antenna	18–24 dB	1.02, 0.95, 0.94	0.97	0.99	1.00	1
Beamforming	14–20 dB	1.90, 1.77, 2.26	1.96	1.93	2.00	2
Alamouti	18–24 dB	1.94, 1.99, 2.61	2.16	1.94	2.00	2
MRC, $L = 2$	16–22 dB	1.83, 2.10, 2.04	2.00	1.96	2.00	2

- The single antenna has diversity order 1. Beamforming, Alamouti and MRC all have diversity order 2.
- In all three two-antenna schemes, the output SNR is proportional to $\|\mathbf{h}\|^2 = |h_1|^2 + |h_2|^2$. This is small only if *both* paths fade at once, which has probability $\approx \mathrm{SNR}^{-2}/2$.
- The Alamouti fit (2.16) is pulled up by its last step (2.61), from 216 to only 65 errors. One 2 dB step with so few errors is accurate to only about ± 0.3.
- Alamouti reaches full diversity even though the transmitter knows nothing about the channel.

SNR gaps (arrows in Fig. 6). Table 6 gives the SNR each scheme needs for a target SER.

Table 6: SNR (dB) needed for a target SER, read off the simulated curves (exact theory in brackets), and the gaps between the schemes. "–": the curve does not reach this SER within 0–24 dB.

	$P_\mathrm{e} = 10^{-2}$	$P_\mathrm{e} = 10^{-3}$	$P_\mathrm{e} = 10^{-4}$
Single antenna	19.46 (19.51)	– (29.58)	– (39.59)
Beamforming	10.15 (10.14)	15.61 (15.60)	20.51 (20.73)
Alamouti	13.15 (13.15)	18.62 (18.61)	23.28 (23.75)
MRC, $L = 2$	10.17 (10.14)	15.59 (15.60)	20.69 (20.73)
Beamforming − MRC	−0.02 (0.00)	+0.02 (0.00)	−0.18 (0.00)
Alamouti − MRC	+2.98 (3.01)	+3.03 (3.01)	+2.60 (3.01)
Alamouti − beamforming	+3.00 (3.01)	+3.01 (3.01)	+2.78 (3.01)
Single − Alamouti	+6.32 (6.36)	– (10.97)	– (15.84)

The 10^{-4} column is the least accurate (about ± 0.3 dB): there the curves are interpolated between points with only 44–216 errors.

- **Beamforming = MRC ($L = 2$): constant 0 dB.** Both give the output SNR $\|\mathbf{h}\|^2\mathrm{SNR}$.
  - With MRC, the receiver collects the signal on two antennas, each with its own independent noise.
  - With beamforming, the transmitter uses its channel knowledge to make the two paths add coherently at the single receive antenna.
  - Both get the same array gain of 2 (3.01 dB, measured 3.01 dB) and diversity 2. The measured differences (≤ 0.18 dB) are Monte Carlo noise. The difference lies in what each needs: MRC needs two receive chains, beamforming needs channel feedback.
- **Alamouti: a constant 3 dB behind both** (exactly $10\log_{10} 2$).
  - Without channel knowledge, the transmitter cannot phase-align the two antennas. It must split its energy equally ($1/\sqrt{2}$ per antenna), and the two signals do not add coherently.
  - The output SNR is $\|\mathbf{h}\|^2\mathrm{SNR}/2$, whose mean equals the SNR: array gain 1 (0 dB, measured 0.00 dB), but full diversity 2.
  - This SNR has exactly the beamforming/MRC distribution scaled by 1/2, so

$$P_\mathrm{e}^{\mathrm{Alamouti}}(\mathrm{SNR}) = P_\mathrm{e}^{\mathrm{BF}}(\mathrm{SNR}/2).$$

    The Alamouti curve is the beamforming curve shifted right by 3.01 dB at *every* SNR, not

only asymptotically. The simulation gives 3.00 and 3.01 dB at 10^{-2} and 10^{-3}.
  - These 3 dB are the price of having no channel knowledge at the transmitter. In return, Alamouti needs no feedback, still sends one symbol per channel use, and is decoded with simple linear processing.
- **Single antenna versus the two-antenna schemes: not constant.** The slopes differ (1 against 2), so the gap grows as the SER falls. Relative to Alamouti it is 6.3 dB at 10^{-2}, and (from exact theory) 11.0 dB at 10^{-3} and 15.8 dB at 10^{-4}. This is a diversity gain, not an array gain: both schemes have array gain 1.

5 Summary of the gaps marked in the figures

Curves compared	Fig.	Gap	Constant? Why
BPSK vs 4-QAM, AWGN	1	3.9 dB (10^{-2}), 3.3 dB (10^{-4})	No; tends to 3 dB (half the energy per component).
BPSK vs 4-QAM, fading	2	≈ 5.6 dB	Yes; parallel slope-1 lines, $10\log_{10}(0.909/0.25)$.
AWGN vs fading, BPSK	3	17.3 dB (10^{-3}), 25.8 dB (10^{-4})	No; exponential vs 1/SNR decay.
MRC vs SC, $L = 1$	4	0 dB	Identical receivers.
MRC vs SC, $L = 2$	4	≈ 1.5 dB	Yes (same slope); array gain 3.01 vs 1.76 dB.
MRC vs SC, $L = 4$	4	3.2–3.5 dB	Yes (same slope); array gain 6.02 vs 3.19 dB.
Beamforming vs MRC ($L = 2$)	6	0 dB	Yes; same output SNR $\|\mathbf{h}\|^2$SNR.
Alamouti vs beamforming / MRC	6	3.0 dB	Yes, exactly; no array gain without channel knowledge.
Single antenna vs Alamouti	6	6.3 dB (10^{-2}), grows	No; diversity 1 vs 2.

In short: two curves with the *same* diversity order are parallel at high SNR, and the constant gap between them is set by array gain or by the modulation. Two curves with *different* diversity orders move further apart as the target SER falls.

References

[1] R. Gangula, *Performance in Narrowband Fading Channels*, Lecture Notes, EC5.407 Wireless Communications, IIIT Hyderabad.

[2] R. Gangula, *Diversity in Narrowband Fading Channels*, Lecture Notes, EC5.407 Wireless Communications, IIIT Hyderabad.

A Numerical results

The tables are read directly from the CSV files in `results/` when the report is compiled. For Problems 2 and 3 they come from the `*_theory.csv` files, which hold the simulated values and the exact theory side by side. "sim." is n_{err}/N with $N = 10^6$. For points with fewer than 10 errors the raw error count is shown instead (in italics); these points are not plotted.

Table 7: Problem 1, AWGN: simulated SER and theory.

	BPSK		4-QAM	
$\mathrm{SNR}_{\mathrm{dB}}$	sim.	theory	sim.	theory
0	$7.93 \cdot 10^{-2}$	$7.86 \cdot 10^{-2}$	$2.93 \cdot 10^{-1}$	$2.92 \cdot 10^{-1}$
1	$5.66 \cdot 10^{-2}$	$5.63 \cdot 10^{-2}$	$2.45 \cdot 10^{-1}$	$2.45 \cdot 10^{-1}$
2	$3.74 \cdot 10^{-2}$	$3.75 \cdot 10^{-2}$	$1.97 \cdot 10^{-1}$	$1.97 \cdot 10^{-1}$
3	$2.33 \cdot 10^{-2}$	$2.29 \cdot 10^{-2}$	$1.52 \cdot 10^{-1}$	$1.52 \cdot 10^{-1}$
4	$1.26 \cdot 10^{-2}$	$1.25 \cdot 10^{-2}$	$1.10 \cdot 10^{-1}$	$1.10 \cdot 10^{-1}$
5	$6.05 \cdot 10^{-3}$	$5.95 \cdot 10^{-3}$	$7.40 \cdot 10^{-2}$	$7.39 \cdot 10^{-2}$
6	$2.43 \cdot 10^{-3}$	$2.39 \cdot 10^{-3}$	$4.54 \cdot 10^{-2}$	$4.55 \cdot 10^{-2}$
7	$7.53 \cdot 10^{-4}$	$7.73 \cdot 10^{-4}$	$2.51 \cdot 10^{-2}$	$2.50 \cdot 10^{-2}$
8	$2.03 \cdot 10^{-4}$	$1.91 \cdot 10^{-4}$	$1.18 \cdot 10^{-2}$	$1.20 \cdot 10^{-2}$
9	$4.00 \cdot 10^{-5}$	$3.36 \cdot 10^{-5}$	$4.80 \cdot 10^{-3}$	$4.82 \cdot 10^{-3}$
10	*6 err.*	$3.87 \cdot 10^{-6}$	$1.54 \cdot 10^{-3}$	$1.56 \cdot 10^{-3}$
11	*0 err.*	$2.61 \cdot 10^{-7}$	$3.68 \cdot 10^{-4}$	$3.88 \cdot 10^{-4}$
12	*0 err.*	$9.01 \cdot 10^{-9}$	$6.30 \cdot 10^{-5}$	$6.86 \cdot 10^{-5}$

Table 8: Problem 1, Rayleigh fading: simulated SER and theory. For 4-QAM both the table approximation and the exact value are listed.

	BPSK		4-QAM		
$\mathrm{SNR}_{\mathrm{dB}}$	sim.	theory	sim.	table approx.	exact
0	$1.47 \cdot 10^{-1}$	$1.46 \cdot 10^{-1}$	$3.65 \cdot 10^{-1}$	$4.23 \cdot 10^{-1}$	$3.65 \cdot 10^{-1}$
2	$1.08 \cdot 10^{-1}$	$1.08 \cdot 10^{-1}$	$2.93 \cdot 10^{-1}$	$3.35 \cdot 10^{-1}$	$2.93 \cdot 10^{-1}$
4	$7.72 \cdot 10^{-2}$	$7.71 \cdot 10^{-2}$	$2.25 \cdot 10^{-1}$	$2.54 \cdot 10^{-1}$	$2.25 \cdot 10^{-1}$
6	$5.37 \cdot 10^{-2}$	$5.30 \cdot 10^{-2}$	$1.64 \cdot 10^{-1}$	$1.84 \cdot 10^{-1}$	$1.64 \cdot 10^{-1}$
8	$3.54 \cdot 10^{-2}$	$3.55 \cdot 10^{-2}$	$1.15 \cdot 10^{-1}$	$1.29 \cdot 10^{-1}$	$1.15 \cdot 10^{-1}$
10	$2.32 \cdot 10^{-2}$	$2.33 \cdot 10^{-2}$	$7.87 \cdot 10^{-2}$	$8.71 \cdot 10^{-2}$	$7.86 \cdot 10^{-2}$
12	$1.49 \cdot 10^{-2}$	$1.51 \cdot 10^{-2}$	$5.20 \cdot 10^{-2}$	$5.77 \cdot 10^{-2}$	$5.22 \cdot 10^{-2}$
14	$9.89 \cdot 10^{-3}$	$9.67 \cdot 10^{-3}$	$3.40 \cdot 10^{-2}$	$3.76 \cdot 10^{-2}$	$3.41 \cdot 10^{-2}$
16	$6.16 \cdot 10^{-3}$	$6.16 \cdot 10^{-3}$	$2.21 \cdot 10^{-2}$	$2.42 \cdot 10^{-2}$	$2.20 \cdot 10^{-2}$
18	$3.96 \cdot 10^{-3}$	$3.92 \cdot 10^{-3}$	$1.42 \cdot 10^{-2}$	$1.55 \cdot 10^{-2}$	$1.41 \cdot 10^{-2}$
20	$2.50 \cdot 10^{-3}$	$2.48 \cdot 10^{-3}$	$8.92 \cdot 10^{-3}$	$9.85 \cdot 10^{-3}$	$8.95 \cdot 10^{-3}$
22	$1.58 \cdot 10^{-3}$	$1.57 \cdot 10^{-3}$	$5.81 \cdot 10^{-3}$	$6.25 \cdot 10^{-3}$	$5.68 \cdot 10^{-3}$
24	$1.01 \cdot 10^{-3}$	$9.92 \cdot 10^{-4}$	$3.60 \cdot 10^{-3}$	$3.96 \cdot 10^{-3}$	$3.60 \cdot 10^{-3}$
26	$6.35 \cdot 10^{-4}$	$6.27 \cdot 10^{-4}$	$2.29 \cdot 10^{-3}$	$2.50 \cdot 10^{-3}$	$2.27 \cdot 10^{-3}$
28	$4.21 \cdot 10^{-4}$	$3.96 \cdot 10^{-4}$	$1.41 \cdot 10^{-3}$	$1.58 \cdot 10^{-3}$	$1.44 \cdot 10^{-3}$
30	$2.84 \cdot 10^{-4}$	$2.50 \cdot 10^{-4}$	$9.13 \cdot 10^{-4}$	$9.99 \cdot 10^{-4}$	$9.08 \cdot 10^{-4}$
32	$1.48 \cdot 10^{-4}$	$1.58 \cdot 10^{-4}$	$5.33 \cdot 10^{-4}$	$6.30 \cdot 10^{-4}$	$5.73 \cdot 10^{-4}$
34	$1.06 \cdot 10^{-4}$	$9.95 \cdot 10^{-5}$	$3.42 \cdot 10^{-4}$	$3.98 \cdot 10^{-4}$	$3.62 \cdot 10^{-4}$
36	$5.70 \cdot 10^{-5}$	$6.28 \cdot 10^{-5}$	$2.25 \cdot 10^{-4}$	$2.51 \cdot 10^{-4}$	$2.28 \cdot 10^{-4}$
38	$4.40 \cdot 10^{-5}$	$3.96 \cdot 10^{-5}$	$1.38 \cdot 10^{-4}$	$1.58 \cdot 10^{-4}$	$1.44 \cdot 10^{-4}$
40	$2.50 \cdot 10^{-5}$	$2.50 \cdot 10^{-5}$	$8.40 \cdot 10^{-5}$	$1.00 \cdot 10^{-4}$	$9.09 \cdot 10^{-5}$

Table 9: Problem 2: simulated 4-QAM SER and exact theory. For $L = 1$, MRC and SC give identical results, equal to the single-antenna columns of Table 10.

	MRC, $L = 2$		SC, $L = 2$		MRC, $L = 4$		SC, $L = 4$	
$\mathrm{SNR}_{\mathrm{dB}}$	sim.	theory	sim.	theory	sim.	theory	sim.	theory
0	$2.10 \cdot 10^{-1}$	$2.10 \cdot 10^{-1}$	$2.64 \cdot 10^{-1}$	$2.64 \cdot 10^{-1}$	$7.70 \cdot 10^{-2}$	$7.73 \cdot 10^{-2}$	$1.82 \cdot 10^{-1}$	$1.82 \cdot 10^{-1}$
2	$1.39 \cdot 10^{-1}$	$1.39 \cdot 10^{-1}$	$1.86 \cdot 10^{-1}$	$1.86 \cdot 10^{-1}$	$3.48 \cdot 10^{-2}$	$3.49 \cdot 10^{-2}$	$1.09 \cdot 10^{-1}$	$1.09 \cdot 10^{-1}$
4	$8.29 \cdot 10^{-2}$	$8.30 \cdot 10^{-2}$	$1.20 \cdot 10^{-1}$	$1.20 \cdot 10^{-1}$	$1.27 \cdot 10^{-2}$	$1.29 \cdot 10^{-2}$	$5.58 \cdot 10^{-2}$	$5.57 \cdot 10^{-2}$
6	$4.54 \cdot 10^{-2}$	$4.52 \cdot 10^{-2}$	$7.02 \cdot 10^{-2}$	$7.01 \cdot 10^{-2}$	$3.83 \cdot 10^{-3}$	$3.94 \cdot 10^{-3}$	$2.39 \cdot 10^{-2}$	$2.38 \cdot 10^{-2}$
8	$2.26 \cdot 10^{-2}$	$2.26 \cdot 10^{-2}$	$3.76 \cdot 10^{-2}$	$3.75 \cdot 10^{-2}$	$1.07 \cdot 10^{-3}$	$1.01 \cdot 10^{-3}$	$8.36 \cdot 10^{-3}$	$8.41 \cdot 10^{-3}$
10	$1.05 \cdot 10^{-2}$	$1.06 \cdot 10^{-2}$	$1.85 \cdot 10^{-2}$	$1.85 \cdot 10^{-2}$	$2.23 \cdot 10^{-4}$	$2.24 \cdot 10^{-4}$	$2.54 \cdot 10^{-3}$	$2.48 \cdot 10^{-3}$
12	$4.76 \cdot 10^{-3}$	$4.69 \cdot 10^{-3}$	$8.69 \cdot 10^{-3}$	$8.56 \cdot 10^{-3}$	$3.90 \cdot 10^{-5}$	$4.48 \cdot 10^{-5}$	$5.95 \cdot 10^{-4}$	$6.23 \cdot 10^{-4}$
14	$2.05 \cdot 10^{-3}$	$2.01 \cdot 10^{-3}$	$3.77 \cdot 10^{-3}$	$3.78 \cdot 10^{-3}$	$1.00 \cdot 10^{-5}$	$8.27 \cdot 10^{-6}$	$1.35 \cdot 10^{-4}$	$1.37 \cdot 10^{-4}$
16	$8.33 \cdot 10^{-4}$	$8.38 \cdot 10^{-4}$	$1.62 \cdot 10^{-3}$	$1.61 \cdot 10^{-3}$	*2 err.*	$1.45 \cdot 10^{-6}$	$2.90 \cdot 10^{-5}$	$2.71 \cdot 10^{-5}$
18	$3.37 \cdot 10^{-4}$	$3.44 \cdot 10^{-4}$	$6.50 \cdot 10^{-4}$	$6.70 \cdot 10^{-4}$	*1 err.*	$2.45 \cdot 10^{-7}$	*5 err.*	$5.00 \cdot 10^{-6}$
20	$1.30 \cdot 10^{-4}$	$1.40 \cdot 10^{-4}$	$2.86 \cdot 10^{-4}$	$2.74 \cdot 10^{-4}$	*0 err.*	$4.04 \cdot 10^{-8}$	*0 err.*	$8.74 \cdot 10^{-7}$
22	$5.20 \cdot 10^{-5}$	$5.62 \cdot 10^{-5}$	$9.40 \cdot 10^{-5}$	$1.11 \cdot 10^{-4}$	*0 err.*	$6.58 \cdot 10^{-9}$	*0 err.*	$1.48 \cdot 10^{-7}$
24	$2.10 \cdot 10^{-5}$	$2.26 \cdot 10^{-5}$	$3.70 \cdot 10^{-5}$	$4.48 \cdot 10^{-5}$	*0 err.*	$1.06 \cdot 10^{-9}$	*0 err.*	$2.44 \cdot 10^{-8}$

Table 10: Problem 3: simulated 4-QAM SER and exact theory.

	Single antenna		Beamforming		Alamouti		MRC, $L = 2$	
$\mathrm{SNR}_{\mathrm{dB}}$	sim.	theory	sim.	theory	sim.	theory	sim.	theory
0	$3.66 \cdot 10^{-1}$	$3.65 \cdot 10^{-1}$	$2.11 \cdot 10^{-1}$	$2.10 \cdot 10^{-1}$	$3.32 \cdot 10^{-1}$	$3.32 \cdot 10^{-1}$	$2.10 \cdot 10^{-1}$	$2.10 \cdot 10^{-1}$
2	$2.93 \cdot 10^{-1}$	$2.93 \cdot 10^{-1}$	$1.39 \cdot 10^{-1}$	$1.39 \cdot 10^{-1}$	$2.51 \cdot 10^{-1}$	$2.50 \cdot 10^{-1}$	$1.39 \cdot 10^{-1}$	$1.39 \cdot 10^{-1}$
4	$2.25 \cdot 10^{-1}$	$2.25 \cdot 10^{-1}$	$8.33 \cdot 10^{-2}$	$8.30 \cdot 10^{-2}$	$1.74 \cdot 10^{-1}$	$1.73 \cdot 10^{-1}$	$8.29 \cdot 10^{-2}$	$8.30 \cdot 10^{-2}$
6	$1.64 \cdot 10^{-1}$	$1.64 \cdot 10^{-1}$	$4.47 \cdot 10^{-2}$	$4.52 \cdot 10^{-2}$	$1.10 \cdot 10^{-1}$	$1.09 \cdot 10^{-1}$	$4.50 \cdot 10^{-2}$	$4.52 \cdot 10^{-2}$
8	$1.16 \cdot 10^{-1}$	$1.15 \cdot 10^{-1}$	$2.24 \cdot 10^{-2}$	$2.26 \cdot 10^{-2}$	$6.19 \cdot 10^{-2}$	$6.21 \cdot 10^{-2}$	$2.26 \cdot 10^{-2}$	$2.26 \cdot 10^{-2}$
10	$7.88 \cdot 10^{-2}$	$7.86 \cdot 10^{-2}$	$1.06 \cdot 10^{-2}$	$1.06 \cdot 10^{-2}$	$3.25 \cdot 10^{-2}$	$3.24 \cdot 10^{-2}$	$1.08 \cdot 10^{-2}$	$1.06 \cdot 10^{-2}$
12	$5.19 \cdot 10^{-2}$	$5.22 \cdot 10^{-2}$	$4.73 \cdot 10^{-3}$	$4.69 \cdot 10^{-3}$	$1.58 \cdot 10^{-2}$	$1.56 \cdot 10^{-2}$	$4.65 \cdot 10^{-3}$	$4.69 \cdot 10^{-3}$
14	$3.42 \cdot 10^{-2}$	$3.41 \cdot 10^{-2}$	$2.03 \cdot 10^{-3}$	$2.01 \cdot 10^{-3}$	$7.13 \cdot 10^{-3}$	$7.11 \cdot 10^{-3}$	$1.94 \cdot 10^{-3}$	$2.01 \cdot 10^{-3}$
16	$2.19 \cdot 10^{-2}$	$2.20 \cdot 10^{-2}$	$8.44 \cdot 10^{-4}$	$8.38 \cdot 10^{-4}$	$3.07 \cdot 10^{-3}$	$3.09 \cdot 10^{-3}$	$8.44 \cdot 10^{-4}$	$8.38 \cdot 10^{-4}$
18	$1.41 \cdot 10^{-2}$	$1.41 \cdot 10^{-2}$	$3.74 \cdot 10^{-4}$	$3.44 \cdot 10^{-4}$	$1.32 \cdot 10^{-3}$	$1.31 \cdot 10^{-3}$	$3.63 \cdot 10^{-4}$	$3.44 \cdot 10^{-4}$
20	$8.82 \cdot 10^{-3}$	$8.95 \cdot 10^{-3}$	$1.32 \cdot 10^{-4}$	$1.40 \cdot 10^{-4}$	$5.39 \cdot 10^{-4}$	$5.40 \cdot 10^{-4}$	$1.38 \cdot 10^{-4}$	$1.40 \cdot 10^{-4}$
22	$5.69 \cdot 10^{-3}$	$5.68 \cdot 10^{-3}$	$4.40 \cdot 10^{-5}$	$5.62 \cdot 10^{-5}$	$2.16 \cdot 10^{-4}$	$2.20 \cdot 10^{-4}$	$5.40 \cdot 10^{-5}$	$5.62 \cdot 10^{-5}$
24	$3.69 \cdot 10^{-3}$	$3.60 \cdot 10^{-3}$	$1.70 \cdot 10^{-5}$	$2.26 \cdot 10^{-5}$	$6.50 \cdot 10^{-5}$	$8.91 \cdot 10^{-5}$	$2.70 \cdot 10^{-5}$	$2.26 \cdot 10^{-5}$